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MAY/JUNE 2013

C A R I B B E A N   E X A M I N A T I O N S   C O U N C I L

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

BIOLOGY

UNIT 1 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULY.

1. This paper consists of SIX questions in two sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question in this booklet.
4. The use of silent non-programmable calculators is allowed.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. (a) Table 1 presents results of an investigation of the nutritional composition of various foods from an average American diet.

TABLE 1: ANALYSIS OF SPECIFIC NUTRIENTS IN COMMON FOODS USING KNOWN REAGENTS

{Note: For reagents which require a colour change to indicate the presence of a nutrient, results were recorded based on a colour index (range) of changes observed for any one reagent.}

Food Substance	FOOD TEST REAGENT			
	Biuret Solution	**Benedict's Solution**	**Lugol's Iodine (KI/I_2) Solution**	**Grease Spot on Brown Paper***
Eggs	Purple	Blue	Black	None seen.
Milk	Pink	Pale yellow	Dark blue	None seen.
Cheerios cereal	Pale blue (almost clear)	Brick-red	Blue-black	Only seen when food sample is dissolved in isopropyl alcohol and then tested.
Hamburger patty	Purple	Blue	Yellow	Clearly seen when rubbed on paper.
Carrot	Pale blue (almost clear)	Orange	Dark blue	None seen.
Potato chips	Pink	Brick-red	Black	Seen when rubbed on paper but not very transparent.
Pepperoni pizza	Purple	Yellow	Dark blue	Clearly seen when rubbed on paper.
Donut	Pink	Orange	Dark blue	Clearly seen when rubbed on paper.

* **Grease Spot on Brown Paper Test:**

1. **Using a glass rod, a sample of the food being tested is rubbed on a section of brown paper (labelled with the sample being tested). "Wet" spot appears on the paper.**

2. **Any excess food which may stick to the paper is removed using a paper towel.**

3. **The sample paper is left to dry for about 10 minutes.**

4. **A translucent spot indicates a positive result for the specific nutrient being tested.**

5. **Isopropyl alcohol serves to dissolve the nutrient in the food substance.**

(i) State the nutrient being tested for, by EACH of the following tests:

Biuret: ______________________________

Benedict's: ______________________________

Lugol's Iodine: ______________________________

Grease spot: ______________________________

[2 marks]

(ii) Using the data presented in Table 1, determine TWO food substances which contain all four nutrients being tested for in this investigation.

____

[1 mark]

(iii) With reference to the qualitative results given in Table 1, suggest which foods are good sources of nutrients for cell structure and growth of the cell, and for supply of energy. Justify your answer with a brief explanation.

Cell structure and growth of the cell:

____

____

____

____

Supply of energy:

____

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____

____

[4 marks]

(b) Figure 1 is a diagrammatic representation of the molecular structure of a short section of a glycogen molecule.

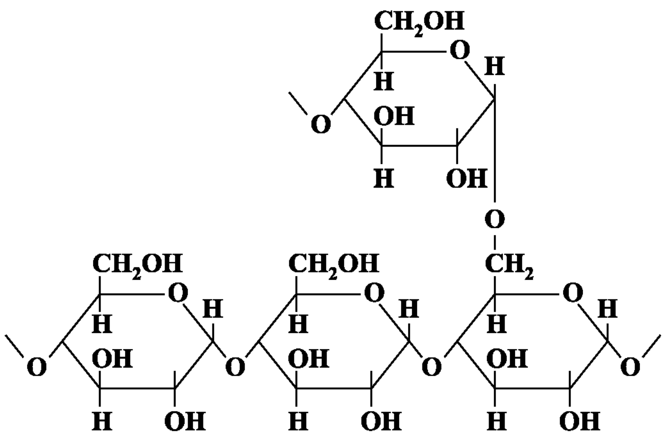

Figure 1. Short section of a glycogen molecule

(i) Using a labelled arrow on Figure 1, highlight the location of EACH of the following:

a) An alpha 1,4-glycosidic linkage

b) An alpha 1,6-glycosidic linkage **[2 marks]**

(ii) In the space below, sketch the general shape of one glycogen molecule.
[Details of individual glucose residues are NOT required.]

Space for diagram

[2 marks]

GO ON TO THE NEXT PAGE

(iii) Amylose (starch) and cellulose are both polymers of glucose molecules but they differ in their structure and function in plant cells. With reference to ONE structural property, briefly explain the functional differences of amylose and cellulose in plant cells.

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[4 marks]

Total 15 marks

2. (a) In a cross between a plant with purple flowers and a plant with white flowers, all the F_1 plants had purple flowers. When the F_1 offspring were crossed (selfed), 705 plants had purple flowers and 224 plants had white flowers.

(i) State the expected ratio for the cross of the F_1 offspring.

____

[1 mark]

(ii) State an appropriate null (H_0) hypothesis and an appropriate alternative (H_1) hypothesis for a Chi-square test of the results.

H_0: ____

____

H_1: ____

____

[2 marks]

(iii) Complete Table 2 by calculating the missing values.

TABLE 2: DATA FOR CHI-SQUARE TEST

Phenotype	**Observed (O)**	**Expected (E)**	**O − E**	**$(O - E)^2/E$**
Purple flowers	705	696		
White flowers	224	233		
$\chi^2 = \Sigma\left[(O - E)^2/E\right]$				

[4 marks]

(iv) Determine the number of degrees of freedom. Show your calculation.

____

[1 mark]

(v) Using the Chi-square values in Table 3, comment on the validity of the null hypothesis stated on page 5.

____

____

____

[2 marks]

TABLE 3: CHI-SQUARE (χ^2) VALUES

Degrees of Freedom	**Number of Classes**	**Chi-square Values**					
1	2	0.46	1.64	2.71	3.84	6.64	10.83
2	3	1.39	3.22	4.61	5.99	9.21	13.82
3	4	2.37	4.64	6.25	7.82	11.34	16.27
4	5	3.36	5.99	7.78	9.49	13.28	18.47
Probability that chance alone could produce this deviation		0.50 (50%)	0.20 (20%)	0.10 (10%)	0.05 (5%)	0.01 (1%)	0.001 (0.1%)

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(b) Figure 2 represents the elongation stage of translation in protein synthesis.

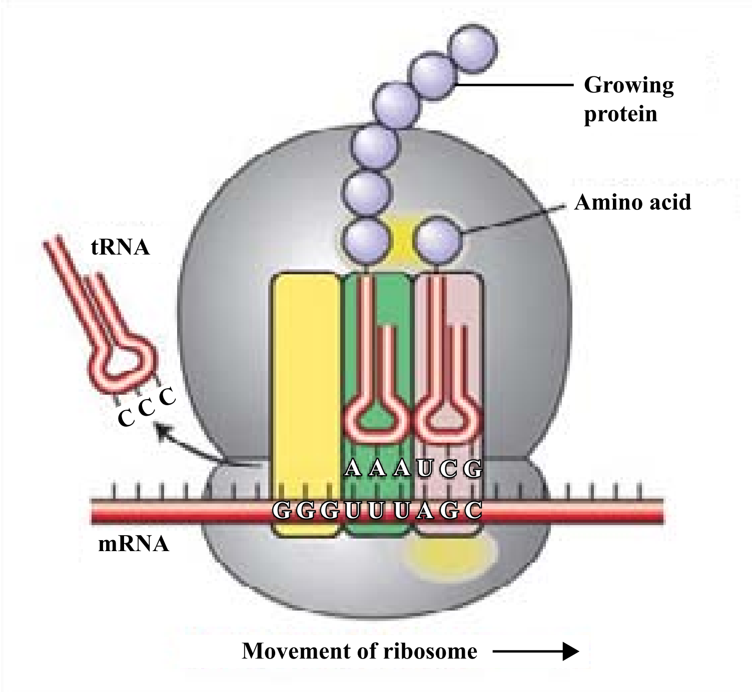

Figure 2. Elongation stage of translation in protein synthesis

Source: http://www.frontiers-in-genetics.org/page.php?id=protein-synthesis_en

(i) What is the term used to describe EACH group of three bases on the

mRNA ______________________________

tRNA? ______________________________

[1 mark]

(ii) Using the mRNA strand below, identify the corresponding triplet bases on the tRNA molecules. Write your answer in the boxes provided. Hint: Begin at the 5′ end.

mRNA **5′ UGG UUU GGC UCA 3′**

[2 marks]

(iii) Outline the nucleotide sequence on the DNA strand which serves as the template for the mRNA strand in (b) (ii).

____

[2 marks]

Total 15 marks

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3. (a) Table 4 is an incomplete table comparing the process of spermatogenesis with that of oogenesis.

TABLE 4: COMPARISION OF THE PROCESS OF SPERMATOGENESIS AND OOGENESIS

Feature	**Spermatogenesis**	**Oogenesis**
Number of meiotic daughter cells which develop into mature gametes		
Duration of the process (mitotic division of stem cells and differentiated spermatogonia/oogonia) in the life span of the individual		

(i) Complete Table 4 with respect to the features listed for spermatogenesis and oogenesis in humans. **[3 marks]**

(ii) Outline the roles of follicle-stimulating hormone (FSH) and luteinizing hormone (LH) in the regulation of spermatogenesis.

FSH: ____

____

____

LH: ____

____

[3 marks]

(b) Figure 3 is a representation of a photomicrograph of a stained section through a human ovary showing developing ova.

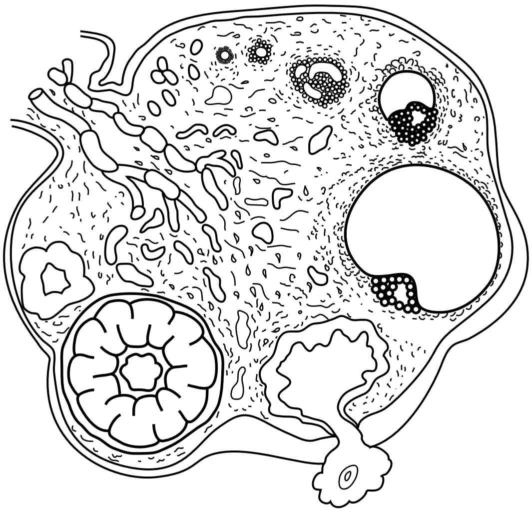

Figure 3. Representation of a photomicrograph of a section through an ovary

In the box below, make a plan drawing of the ovary to include the stage of the developing ovum just prior to release from the ovary.

[6 marks]

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(c) Figure 4 is a diagram of a section through a carpel of an angiosperm plant showing the process of double fertilization.

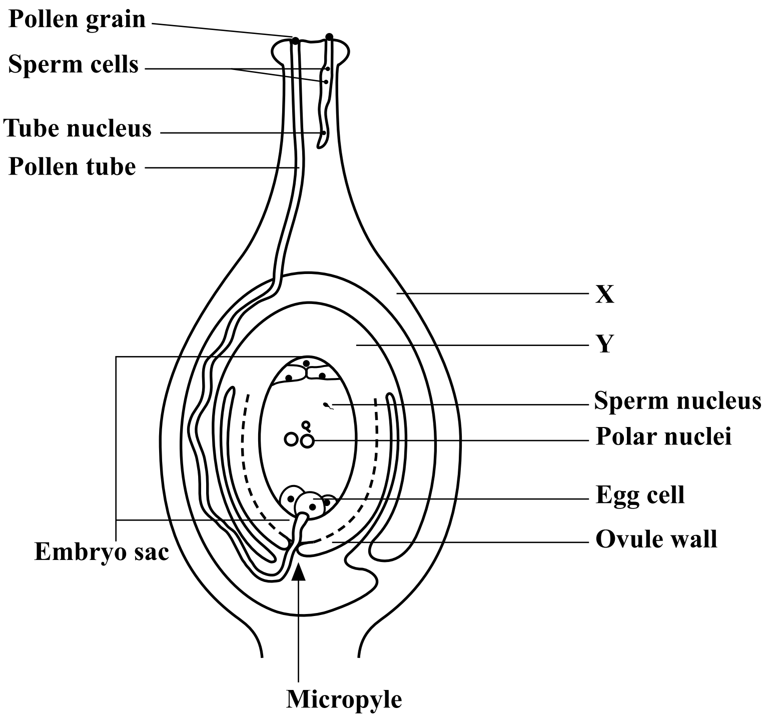

Figure 4. Section through the carpel of an angiosperm plant

Comment on the fate of the structure labelled X and the structure labelled Y, which includes the ovule walls and the embryo sac and its contents.

X: ____

____

Y: ____

____

____

____

[3 marks]

Total 15 marks

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SECTION B

Answer ALL questions.

Write your answers in the spaces provided at the end of each question.

4. (a) Define the term 'diffusion', and distinguish between active transport and facilitated diffusion in cells. **[4 marks]**

(b) (i) With the aid of a simplified labelled diagram, describe the fluid mosaic model of membrane structure. **[6 marks]**

(ii) With reference to the fluid mosaic model, discuss how polar molecules and ions are transported. **[5 marks]**

Total 15 marks

Write the answer to Question 4 here.

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Write the answer to Question 4 here.

Space for diagram.

5. (a) *Mycobacterium tuberculosis,* the bacterium that causes tuberculosis, has been virtually eliminated by the widespread use of two effective antibiotics. Due to the development of antibiotic-resistant strains of the bacterium, the disease has re-emerged as a major public health problem.

(i) Applying Darwin's theory of evolution by natural selection, discuss how resistance to antibiotics could have evolved in bacteria. Include in your discussion a concise explanation of natural selection. **[6 marks]**

(ii) Briefly comment on why mutation is regarded as a driving force of evolution. **[2 marks]**

(b) Define the term 'speciation', and using examples, explain how geographical isolation may lead to speciation. **[7 marks]**

Total 15 marks

Write the answer to Question 5 here.

Write the answer to Question 5 here.

6. (a) (i) Give a concise description of the key stages in the fertilization of a human secondary oocyte by a spermatozoon. Begin your account with the acrosome reaction. **[5 marks]**

(ii) Briefly comment on the significance of the process of fertilization. **[2 marks]**

(b) (i) Describe TWO major functions of the placenta. **[4 marks]**

(ii) Discuss TWO ways in which maternal behaviour can affect foetal development. **[4 marks]**

Total 15 marks

Write the answer to Question 6 here.

Write the answer to Question 6 here.

Write the answer to Question 6 here.

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.